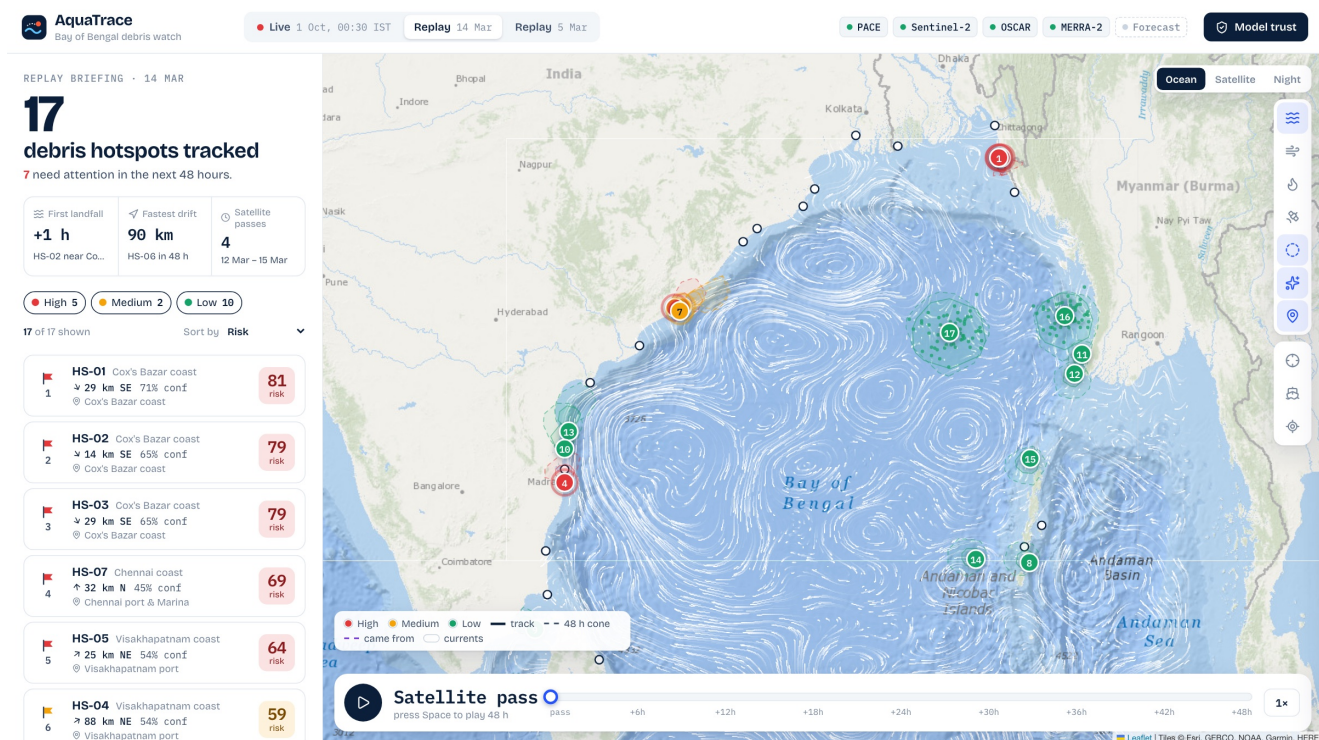


AquaTrace AI

From detecting marine debris to predicting its next move: satellite detection, 24–48 hour drift forecasting and cleanup prioritisation for the Bay of Bengal



The AquaTrace AI dashboard: 17 hotspots from the 14 March 2026 NASA replay over animated ocean currents, ranked by cleanup risk.

94.1% detection accuracy (15 classes, unseen test data)	0.93 F1 score for floating material	16.6 km median forecast error at +24 h (30.4 km at +48 h), 87 held-out buoy tests	59% more accurate than assuming debris stays where seen
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Date: 01 October 2026 · **Region:** Bay of Bengal and Andaman Sea (4°–23°N, 79°–96°E) · **Data:** NASA PACE, OSCAR, MERRA-2; ESA Sentinel-2; MARIDA; NOAA drifters

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1. Executive summary

Marine debris does not stay where a satellite sees it. Currents and wind keep moving it, cleanup boats are few, and a daily satellite image is only a snapshot. **AquaTrace AI** turns free NASA and ESA satellite data into a ranked, forecast-aware cleanup plan for the Bay of Bengal. It answers four questions: *where is floating debris likely now, where will it be in 24–48 hours, where did it come from, and how should a boat reach it?*

The system runs as a pipeline with six stages: **Observe → Detect → Fuse → Forecast → Prioritize → Act**. It reads hyperspectral imagery from NASA's PACE satellite and higher-resolution Sentinel-2 imagery near the coast. Floating material is flagged with a spectral index (the Floating Debris Index) plus an AI classifier trained on the MARIDA marine-debris benchmark. Each hotspot is moved forward with a 300-particle Lagrangian drift model driven by NASA OSCAR currents and MERRA-2 wind (or forecast currents and wind in live mode). It is then scored for cleanup risk and shown on an interactive map with recommended actions and boat routes.

Key results

Result	Value	How it was measured
Detection accuracy (15 classes)	94.1%	MARIDA official test split, 194,863 unseen labelled pixels
F1, floating material	0.932	precision 0.90, recall 0.96
F1, marine debris	0.781	precision 0.70, recall 0.89
Confidence calibration error	0.9%	expected calibration error, floating material
Drift forecast error, +24 h	16.6 km	median of 87 forecasts vs real NOAA buoys (held-out month)
Drift forecast error, +48 h	30.4 km	vs 73.8 km for a static snapshot
Skill over a snapshot, +48 h	59%	$1 - (\text{model error} / \text{persistence error})$
Real buoy inside the 90% cone	79.3% / 86.2%	+24 h / +48 h, held-out month (before: 5.7% / 4.6%)
Wind drag (windage)	2–3%	tuned on January 2025 buoys (lowest error at 2.5%)

What is new

- **Closes the whole loop:** detection, 48-hour forecast, risk ranking and a boat route in one tool.
- **Near-real-time nowcasting:** detections from the latest passes (online about 3 hours after each daily PACE pass) are moved to their likely position *now*, so the map stays useful between passes and under clouds.
- **Source tracking:** the drift model runs backwards 72 hours to point at likely river or coastal sources.
- **Moving-target cleanup routes:** boats are sent to where each patch *will be* when they arrive.
- **One AI model for two satellites:** background-relative features let a model trained on Sentinel-2 labels run on PACE.
- **Two-expert classifier:** Extra Trees decides *floating material or not*; a vote with gradient boosting names everything else. Accuracy rose from 91.9% to 94.1% without losing debris detection.
- **Honest uncertainty:** the 90% forecast cone is checked against real buoys. It used to contain them 5% of the time; with a per-particle current error it now contains them 79–86% of the time on a month of buoys not used for tuning.

2. Problem and objectives

2.1 Problem statement

More than 11 million tonnes of plastic enter the ocean every year (UNEP, 2021). Most comes from land through rivers and coastal cities, and the Bay of Bengal receives the outflow of some of the world's largest river systems (Ganga–Brahmaputra–Meghna, Godavari, Krishna, Mahanadi, Irrawaddy). Monitoring it is hard for three reasons:

- The ocean is far too large for continuous manual or ship-based monitoring.
- Satellite observations are snapshots. They show where floating material was, not where it is going.
- Surface currents and wind move floating material tens of kilometres a day, and cleanup resources are limited, so priority zones must be chosen in advance.

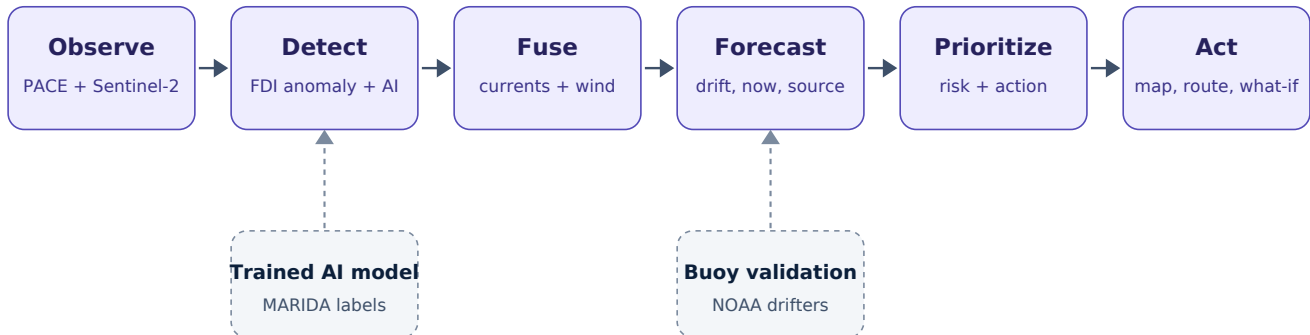
Core question: how can we detect potential floating-debris hotspots early and forecast where they will move, so cleanup teams can act before debris reaches sensitive coasts?

2.2 Objectives and status

#	Objective	Status
1	Detect potential floating-debris hotspots from NASA PACE OCI surface reflectance, with a confidence score	Done (plus Sentinel-2 coastal layer)
2	Forecast hotspot movement over 24–48 h by fusing NASA OSCAR currents and MERRA-2 wind	Done, validated on held-out buoy tracks
3	Compute a cleanup risk score from confidence, predicted location, recurrence, proximity to sensitive areas and forecast uncertainty	Done
4	Deliver an interactive cleanup-priority map (FastAPI + React + Leaflet)	Done (redesigned dashboard)
5	Measure detection precision/recall/F1, 24/48 h forecast error and confidence calibration	Done
+	Near-real-time live mode, source back-tracking, moving-target route planner, what-if simulator	Added beyond the original plan

3. System overview

AquaTrace AI is a Python pipeline that writes one self-contained result file per run, plus a web dashboard that displays it. The figure shows the six stages of a run and the two components built once, offline.



Solid: one pipeline run. Dashed: built once, offline (training and validation).

Figure 1. The AquaTrace AI workflow.

Stage	What happens	Output	Code
Observe	Download the latest PACE pass (whole Bay, ~1.2 km) and Sentinel-2 scenes (20 m, 5 coastal sites); mask land, cloud, glint	Clean reflectance scenes	<code>sources/nasa.py</code> , <code>sources/sentinel2.py</code>
Detect	Floating Debris Index anomaly against local water background; AI classifier rejects ships, wakes, waves, clouds; group pixels	Hotspots with confidence	<code>detect.py</code> , <code>marida.py</code>
Fuse	Load currents and 10 m wind on a common grid (NASA OSCAR + MERRA-2, or Open-Meteo forecasts for live runs)	Velocity fields	<code>forcing.py</code> , <code>sources/</code>
Forecast	300-particle Lagrangian ensemble; nowcast to the present (live); 72 h back-track to the source	Tracks, cones, beaching, source	<code>drift.py</code> , <code>source.py</code>
Prioritize	Weighted risk score and tier; plain-language recommended action	Ranked list	<code>risk.py</code>
Act	Interactive map, what-if simulator and moving-target vessel routes	Dashboard	<code>api.py</code> , <code>route.py</code> , <code>frontend/</code>

3.1 Two ways to run

Mode	Command	Satellite data	Currents and wind	Use
Live	<code>pipeline --live</code>	Best 3 PACE near-real-time passes from the last 3 days + latest Sentinel-2	Open-Meteo forecasts (Mercator SMOC currents, GFS/ECMWF wind)	Today's map, forecast from now
Replay	<code>pipeline --date YYYY-MM-DD</code>	PACE standard product + Sentinel-2 around that date	NASA OSCAR v2.0 + MERRA-2	Proof: compare with what happened

3.2 Daily operation (live mode)

- ~12:00–13:00 IST: PACE passes over the Bay of Bengal.
- ~3 hours later: NASA publishes the near-real-time Level-2 product (for example, the 06:26 UTC pass on 30 September 2026 was online at 09:11 UTC).
- ~16:00 IST: run `pipeline --live`: about 9 minutes of processing, plus the time to download ~2 GB of PACE data (around 10–15 minutes on a typical connection).

- The dashboard checks for new runs every 5 minutes and switches to the newest live run automatically.
- On cloudy days the nowcast keeps earlier detections moving, so the map never goes blank.

4. Data sources

All datasets are free. Only the three NASA products need an (also free) Earthdata login. The table summarises them; details follow.

Dataset	Provider	Resolution	Used for
PACE OCI L2 SFREFL (standard + NRT), v3.1	NASA Ocean Biology DAAC	~1.2 km, 122 bands, daily	Regional detection across the Bay
Sentinel-2 L2A	ESA Copernicus (via Earth Search STAC)	10–20 m, every 2–5 days	Coastal detection at 5 sites
OSCAR v2.0 (final / interim / NRT)	NASA PO.DAAC	0.25°, daily	Ocean currents (replays, validation)
MERRA-2 M2T1NXSLV v5.12.4	NASA GES DISC (GMAO)	0.5° × 0.625°, hourly	10 m wind (replays, validation)
MARIDA	Kikaki et al. (2022), Zenodo	Sentinel-2, 10 m, 15 classes	Training and testing the AI model
Global Drifter Program, 6-hourly QC	NOAA AOML (ERDDAP)	buoy positions every 6 h	Validating the drift forecast
Open-Meteo Marine / Archive / Forecast APIs	Open-Meteo (Mercator SMOC, ERA5, GFS/ECMWF)	sampled on a 0.5° grid, hourly	Live forecasts; no-login fallback
Natural Earth 10 m land + coastline	Natural Earth	1:10 million	Land mask, beaching, coast distance, routing

4.1 NASA PACE OCI surface reflectance

PACE (Plankton, Aerosol, Cloud, ocean Ecosystem), launched in February 2024, carries the Ocean Color Instrument (OCI), a hyperspectral imager with near-daily coverage. We use the Level-2 surface-reflectance product PACE_OCI_L2_SFREFL (and its near-real-time twin _NRT, version 3.1).

Property	Value
Bands	122: 346–895 nm (hyperspectral) plus SWIR at 1038, 1249, 1618, 2131 and 2258 nm
Pixel size	about 1.2 km at nadir; granule of 1710 lines × 1272 pixels
File size	about 730 MB per 5-minute granule (netCDF-4)
Variables read	geophysical_data/rhos, geophysical_data/l2_flags, navigation_data/latitude and longitude, sensor_band_parameters/wavelength_3d
Quality flags masked	LAND, CLDICE, HIGLINT, STRAYLIGHT, ATMFAIL, HILT, NAVFAIL, PRODFAIL, HISATZEN, HISOLZEN
Granule selection	ranked by (share of the Bay covered) × (1 – cloud cover)
Passes used	Replay: 14 Mar 2026 06:59 UTC (56% cloud). Live: 28 Sep 06:55, 29 Sep 07:29 and 30 Sep 06:26 UTC (82%, 69%, 95% cloud)
Access	earthaccess (Python) with an Earthdata login stored in ~/.netrc

4.2 ESA Sentinel-2 L2A

Sentinel-2's Multispectral Instrument provides 13 bands at 10, 20 and 60 m. We read 11 bands (B1–B8A, B11, B12) and the scene classification layer (SCL) as cloud-optimised GeoTIFFs from the open Earth Search STAC catalogue. Each site is read as a 30 × 30 km window at 20 m (1500 × 1500 pixels), choosing the pass closest in time within ± 6 days with under 20% cloud. Water pixels come from SCL class 6; clouds, cirrus and shadows (classes 3, 8, 9, 10) are masked with a 5-pixel buffer. Note: when `earthsearch:boa_offset_applied` is true, the -0.1 offset must not be added again (doing so gives negative reflectance).

Coastal sites: Chennai, Hooghly estuary / Sagar Island, Visakhapatnam, Cox's Bazar and Port Blair (Andaman).

4.3 NASA OSCAR v2.0 surface currents

OSCAR (Ocean Surface Current Analyses Real-time) combines satellite sea-surface height, wind and temperature into daily 0.25° surface currents (u , v in m/s). Three versions are searched in order of quality: final (available until January 2026), interim and near-real-time. The final version was used for the January–February 2025 buoy validation and the interim version for the March 2026 replays. Each daily file (~ 33 MB, global) is subset to the Bay; daily means are time-stamped at 12:00 UTC.

4.4 NASA MERRA-2 10 m wind

MERRA-2 is NASA GMAO's atmospheric reanalysis. From the hourly single-level collection M2T1NXSLV we read the 10 m wind components U10M and V10M on a $0.5^\circ \times 0.625^\circ$ grid. Only the Bay of Bengal subset is requested through NASA's cloud OPeNDAP service (DAP4 constraint expressions), about 240 KB per day instead of the 396 MB global daily file.

4.5 MARIDA (Marine Debris Archive)

MARIDA is a public benchmark of Sentinel-2 image patches (256×256 pixels at 10 m, 11 bands) with pixel labels in 15 classes and an annotator confidence level (high, moderate, low). It has official splits of 694 training, 328 validation and 359 test patches (1,381 in total). The pixel counts below are for training plus validation.

Class	Pixels	Class	Pixels	Class	Pixels
Marine Debris	3,018	Ship	4,629	Sediment-Laden Water	279,900
Dense Sargassum	2,037	Clouds	84,557	Foam	838
Sparse Sargassum	1,476	Marine Water	105,716	Turbid Water	125,386
Natural Organic Material	815	Cloud Shadows	8,079	Shallow Water	14,863
Waves	3,962	Wakes	6,920	Mixed Water	318

4.6 NOAA Global Drifter Program

Surface drifting buoys tracked by satellite. We use the quality-controlled, 6-hourly interpolated dataset (`drifter_6hour_qc`, NOAA AOML ERDDAP), available until mid-2025. Seven drifters were in the Bay during January–June 2025. Five were active over 2025-02-01 to 2025-02-24, giving 87 buoy-day test cases (each with a known position at the start, +24 h and +48 h). All five had lost their drogue, so they float at the surface and are closer in behaviour to floating debris.

4.7 Open-Meteo (live forecasts and fallback)

NASA OSCAR and MERRA-2 are published days to weeks after the fact, so live runs use forecasts from Open-Meteo: the Marine API (Mercator / Météo-France SMOC ocean model currents, hourly), the Forecast API (GFS/ECMWF 10 m wind) and the Archive API (ERA5 reanalysis wind for past dates). Points are requested on a 0.5° grid (about 1,365 points over the Bay), cached as netCDF, and converted from speed and direction to u/v components.

4.8 Natural Earth and curated reference lists

Natural Earth 10 m land polygons and coastlines provide the land mask (beaching, inland-water removal, route checks) and distance to the coast. Three hand-entered lists with approximate coordinates support the analysis: **19 sensitive coastal sites** (mangroves such as the Sundarbans and Bhitarkanika, olive-ridley nesting beaches, Ramsar lagoons, Andaman coral reefs and major ports), **12 river mouths** for source attribution, and **8 port anchorages** for route planning.

5. Methods: how each model works

This section describes every processing step and model with its parameters, exactly as implemented. Settings live in `backend/aquatrace/config.py` and the module named in each heading.

5.1 Pre-processing and masking

One spectral language for two sensors. PACE is hyperspectral, so each Sentinel-2-equivalent band is the mean of the PACE bands inside that band's wavelength range. The detector and classifier then see the same 11 bands from either satellite.

Sentinel-2 band	B1	B2	B3	B4	B5	B6
Range (nm)	433-453	458-523	543-578	650-680	698-713	733-748
Sentinel-2 band	B7	B8	B8A	B11	B12	
Range (nm)	773-793	785-899	855-875	1565-1660	2100-2280	

- **PACE masks:** the L2 quality flags listed in 4.1; an extra haze test marks pixels with B8A reflectance above 0.06 as cloud; the cloud mask is grown by 3 pixels because cloud edges cause most false alarms.
- **Sentinel-2 masks:** only SCL water (class 6); clouds, cirrus and shadows are removed with a 5-pixel buffer.
- **Candidate exclusions:** very bright objects (B2 more than 0.06 above the local background, or B11 above 0.10: ships, platforms, cloud fragments); a shoreline strip (2.5 km for PACE, 0.4 km for Sentinel-2) where mixed land/water pixels mimic floating material; and inland water (backwaters, lakes), since this is a *marine* product.

5.2 Spectral anomaly detection (detect.py)

Floating material raises near-infrared reflectance above the dark water around it. We use the Floating Debris Index (Biermann et al., 2020) and, as diagnostics, the Floating Algae Index (Hu, 2009) and NDVI:

```
FDI = R(B8) - [ R(B6) + (R(B11) - R(B6)) * ((λ8 - λ4) / (λ11 - λ4)) * 10 ]
FAI = R(B8) - [ R(B4) + (R(B11) - R(B4)) * (λ8 - λ4) / (λ11 - λ4) ]
NDVI = (R(B8) - R(B4)) / (R(B8) + R(B4))
band centres: λ4 = 664.8, λ6 = 740.5, λ8 = 832.8, λ11 = 1612 nm
```

Each pixel's FDI is compared with its local open-water background, the median of valid pixels in blocks of 41 pixels (PACE, about 50 km) or 101 pixels (Sentinel-2 at 20 m, about 2 km), upsampled smoothly. The difference becomes a robust z-score using the median absolute deviation (MAD):

$$z = (\text{FDI} - \text{background}) / (1.4826 \times \text{MAD})$$

candidate pixel: $z \geq 3.5$ and FDI above its background

5.3 AI classifier (marida.py)

A pixel classifier separates real floating material from look-alikes (ships, wakes, waves, clouds, turbid or shallow water). It uses 44 features per pixel:

- **15 background-relative spectral features:** the 11 bands minus the local water background, FDI and FAI of those differences, and normalised $\Delta\text{NDVI} = (\Delta\text{B8} - \Delta\text{B4}) / (|\Delta\text{B8}| + |\Delta\text{B4}| + 0.001)$ and $\Delta\text{NDWI} = (\Delta\text{B3} - \Delta\text{B8}) / (|\Delta\text{B3}| + |\Delta\text{B8}| + 0.001)$. Subtracting the background removes most of the difference between processing levels (MARIDA vs Sentinel-2 L2A vs PACE).
- **29 texture features:** for ΔFDI , ΔNDVI , ΔB8 and ΔB2 , the mean and standard deviation in 3×3 and 7×7 windows, plus local contrast (value – 7×7 mean); added in round 2: the 15×15 mean and standard deviation of ΔFDI and ΔNDVI , edge strength (Sobel gradient) of ΔFDI , ΔB8 and ΔB2 , and the 7×7 standard deviation of ΔB4 and ΔB11 . Debris lines, ships, waves and foam differ in shape as much as in colour; windrows are thin lines with sharp edges.

Setting	Value
Algorithm	Two experts (scikit-learn). Extra Trees (extremely randomised trees; Geurts et al., 2006) decides floating material or not; a 50/50 soft vote of Extra Trees and histogram gradient boosting names every other class
Settings	Extra Trees: 500 trees, minimum 2 samples per leaf. Gradient boosting: 400 rounds, learning rate 0.08, 63 leaves, L2 regularisation 1.0
Class imbalance	balanced class weights (per bootstrap sample for Extra Trees); at most 40,000 training pixels per class
Training data	MARIDA train + validation splits (choices made on validation only; see 6.1)
Evaluation	MARIDA test split, scored once per development round
At inference	pixels classed as Ship, Clouds, Waves, Cloud Shadows or Wakes are rejected; $P(\text{floating}) = P(\text{Marine Debris}) + P(\text{Dense Sargassum}) + P(\text{Sparse Sargassum})$ from Extra Trees; on Sentinel-2 a candidate needs $P(\text{floating}) \geq 0.5$

5.4 Hotspots and confidence

Candidate pixels are grouped into hotspots (8-connected components after a 1-pixel dilation; at least 2 pixels). Each hotspot gets a confidence between 0 and 1:

```
A = clip( log10(mean z / 3.5) / 1.5, 0, 1 )      anomaly strength (log scale)
Sz = clip( ln(1 + pixels) / ln(41), 0, 1 )      hotspot size
Sentinel-2: conf = 0.45·A + 0.15·Sz + 0.40·P(floating)
PACE:       conf = 0.65·A + 0.15·Sz + 0.20·P(floating)
            (a 10 m-trained model on ~1 km pixels is supporting evidence only)
then        conf × (1 – 0.5 × cloud share)
            cloud share = part of the hotspot plus a 4-pixel margin that is cloud or invalid
```

A run keeps the 10 most confident hotspots per PACE scene and 3 per Sentinel-2 site (25 at most). Hotspot radius is $\max(\text{pixel size}, \sqrt{(\text{area} / \pi)})$.

5.5 Fusing currents and wind (forcing.py)

Every current or wind source is converted to one structure: u and v (m/s, east and north) on a (time, latitude, longitude) grid. Land gaps are filled with the nearest valid value so interpolation near the coast stays finite. The land polygon, not the grid, decides beaching. Values are interpolated linearly in time and space. Outside the time window the nearest time is used (persistence). Replays load 3 days before to 3 days after the pass. Live runs load 4 days before the oldest pass to 3 days after now, which allows back-tracking.

5.6 Lagrangian drift model (drift.py)

Each hotspot is released as an ensemble of 300 virtual particles, spread in a Gaussian cloud of the hotspot's radius (at least 0.5 km), plus one central particle with average windage and no randomness. Every hour each particle moves with:

```
dx/dt = u_current(x, t) +  $\alpha$  · u_wind10m(x, t) + e      (4th-order Runge-Kutta,  $\Delta t = 1$  h)
      + random walk  $N(0, 1) \cdot \sqrt{2 K \Delta t}$        $K = 20 \text{ m}^2/\text{s} \rightarrow \sigma \approx 380 \text{ m per step}$ 
 $\alpha \sim \text{Uniform}(2\%, 3\%)$  per particle (unknown debris type)
 $e \sim N(0, 0.16 \text{ m/s})$  per particle and axis, constant in time, in  $\pm$  pairs (current error)
 $\Delta \text{lat} = \Delta y / 111,320 \text{ m}, \quad \Delta \text{lon} = \Delta x / (111,320 \text{ m} \cdot \cos \text{lat})$ 
```

- **Beaching:** a particle that moves from sea onto land (Natural Earth 10 m) stops and is counted as beached. Particles that start inside the coarse land polygon near the shore are not counted at $t = 0$.
- **Outputs:** the hourly central track; particle snapshots every 3 h; and at +24 h and +48 h the mean position, spread (root-mean-square distance from the mean), distance travelled, share beached and a 90% uncertainty polygon (convex hull of the 90% of particles nearest the median), plus the time by which half the particles are ashore.
- **Current error:** daily 25 km current maps miss tides, inertial motion and small eddies. Against real buoys the forecast error grows in proportion to time, the signature of a velocity error, which diffusion (growing with $\sqrt{t}$) cannot reproduce. Each particle therefore carries its own constant current error; the $\pm$ pairs widen the cone without moving its centre.
- **Tuned on a held-out design:** windage and current error were chosen on January 2025 buoys and scored on February 2025 (6.3). Windage 2.5% gave the lowest January error, so the ensemble samples 2–3%.

5.7 Nowcasting (live mode)

A detection seen h hours ago is drifted forward h hours to the issue time. Its new position is the mean of the particles still afloat. If 80% or more have beached, the hotspot is flagged as *likely already ashore* and the recommendation becomes a beach survey. Its radius grows to the nowcast spread, and the 48-hour forecast then starts from *now*.

5.8 Source back-tracking (source.py)

The same particle model runs with time reversed for 72 hours. Back-tracked particles that reach land mark a coastal source. Each is assigned to the nearest of 12 river mouths if one is within 60 km. If at least 30% of particles reach the coast, the summary names the dominant river mouth (land-based source) or *local coastline*. Otherwise it reports where the patch was 72 hours earlier, at sea.

5.9 Recurrence

In live mode, detections from different passes within 15 km of each other are merged and the newest is kept. Recurrence = (passes in which it was seen – 1) / (passes – 1). Replays compare with hotspots from earlier runs.

5.10 Cleanup risk score (risk.py)

```

risk = 100 × ( 0.35·confidence + 0.25·proximity + 0.15·beaching
              + 0.15·recurrence + 0.10·certainty )
proximity = 0.75 · max over sites( weight · e-(dsite / 60 km) )
              + 0.25 · e-(dcoast / 20 km)          (closest along the 48 h track)
beaching = share of particles ashore by +48 h
certainty = 1 - min( flow spread at 48 h / 25 km, 1 )
              flow spread = spread without the current-error and diffusion terms
              (those are the same for every hotspot, so they carry no ranking information)
tiers: High ≥ 60, Medium ≥ 40, Low < 40

```

Site weights are 0.6–1.0: 1.0 for the Sundarbans, Bhitarkanika–Gahirmatha, Mahatma Gandhi Marine National Park and St. Martin's Island; 0.8–0.9 for other mangroves, lagoons and reefs; 0.6–0.7 for ports and busy beaches. Recommended actions follow simple rules:

- Nowcast says ashore → send a beach cleanup or survey team.
- Half of the particles ashore within 24 h → send a shoreline team and alert the nearest sensitive site.
- High tier → deploy a vessel within 24 h and intercept at the forecast +24 h position.
- Medium tier → schedule a patrol and re-check the next satellite pass. Low tier → monitor.

5.11 Moving-target route planner (route.py)

From a port anchorage, the planner repeatedly picks the next hotspot that minimises *travel time* × $(1.6 - \text{risk}/100)$, favouring hotspots that are close *and* high-risk. Debris keeps moving while the boat travels, so each stop is aimed at the hotspot's forecast position at the boat's arrival time, found by fixed-point iteration (6 steps) on the forecast track. Defaults: 10 knots, 2 hours of cleanup per stop, candidates within 450 km. A leg that would cross land detours through the shortest single offshore waypoint (offsets of 10–110 km at 25%, 50% and 75% of the leg). Stops beyond the 48 h forecast or already ashore are skipped.

6. Accuracy and validation

6.1 Detection: the AI classifier

The final model was trained on MARIDA's training and validation splits and scored on the official test split (194,863 labelled pixels it never saw). All model and feature choices were made on the validation split. The test split was scored once per development round (twice in total), and the round-2 model was kept whatever its test score, so the numbers are not tuned to the test data.

94.1% overall accuracy, 15 classes	0.776 macro F1 (all classes equal)	0.932 floating-material F1 (P 0.90, R 0.96)	0.781 marine-debris F1 (P 0.70, R 0.89)
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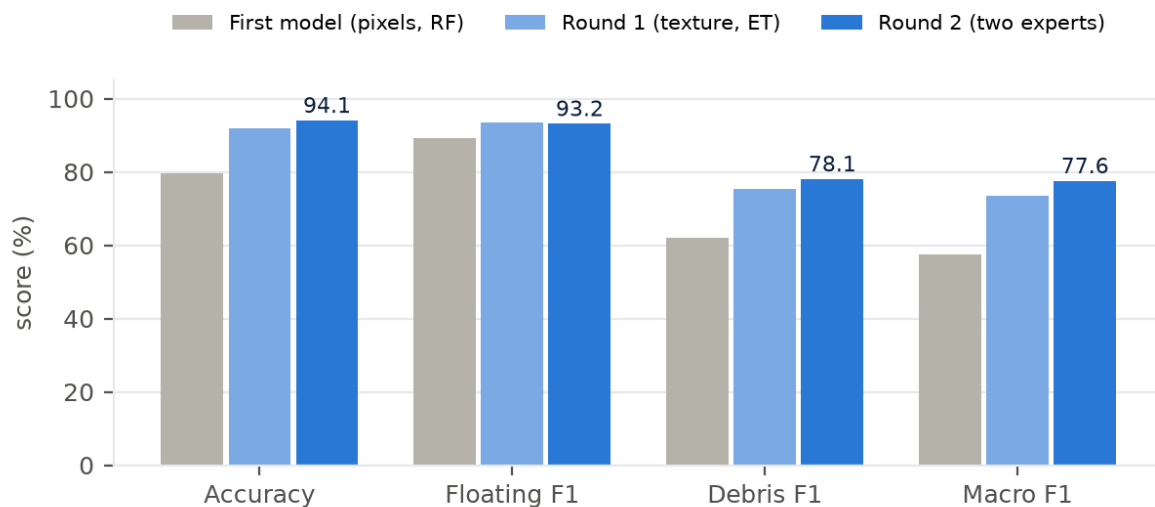
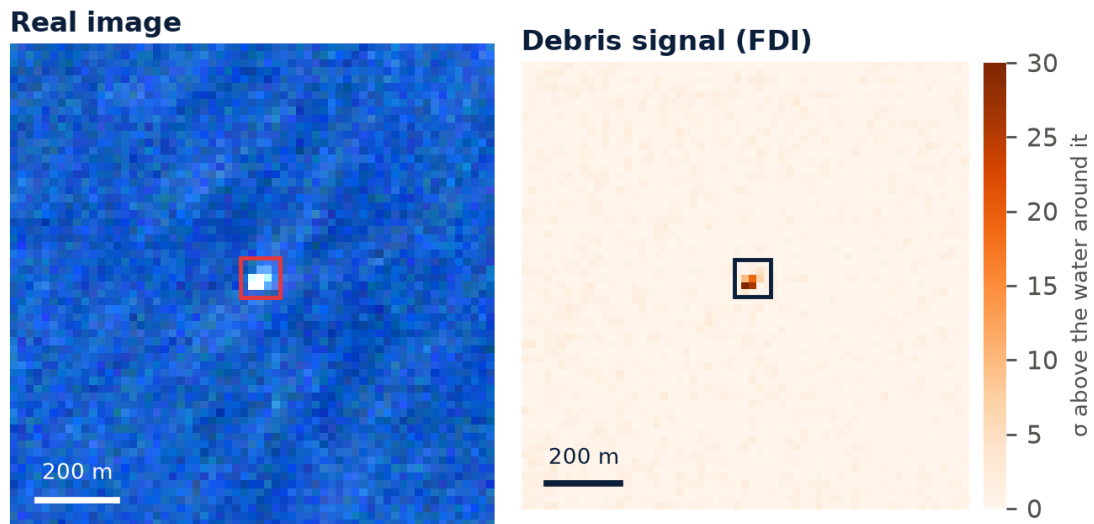


Figure 2. First model (pixel features, Random Forest), round 1 (texture features, Extra Trees) and round 2 (current: two experts, features v2) on the MARIDA test split.

How to read these numbers. Most labelled pixels are water or cloud, so overall accuracy is flattered by easy classes. The F1 scores for the classes that matter are more informative. *Recall* 0.89 for debris means the model finds 89 of every 100 debris pixels; *precision* 0.70 means about 7 in 10 pixels it calls debris really are. In the full pipeline, clouds are masked and ships, wakes and waves are rejected before the model's vote counts, so operational false alarms are lower than the raw precision suggests.

Evidence: how a satellite can see floating debris

No satellite can see a bottle: Sentinel-2 pixels are 20 m (10 m for some bands) and PACE pixels about 1.2 km. What a satellite measures is how much sunlight each pixel reflects in each colour band. A patch or line of floating material changes that light in a recognisable way. The clear map in the dashboard is a background basemap, and the dots on it are forecast particles; the detections themselves come from pixels like the ones below.



Sentinel-2, 2026-03-12 05:03 UTC · 20 m pixels · boxed: 6 flagged ($\approx 2,400 \text{ m}^2$)

Figure 2b. Hotspot HS-04, 9 km off Visakhapatnam (Sentinel-2, 2026-03-12): 6 pixels stand out from the water around them (up to 33 robust standard deviations).

The physics. Water absorbs near-infrared light, so open sea is dark in the near-infrared bands. Anything floating on the surface reflects near-infrared light back. The Floating Debris Index (FDI; Biermann et al., 2020) measures that excess against a baseline drawn between the red-edge and short-wave infrared bands. The shape of the reflection across bands is a fingerprint: floating algae and vegetation jump sharply at the red edge (chlorophyll); plastic debris is flatter and bright; ships are bright even in the short-wave infrared.

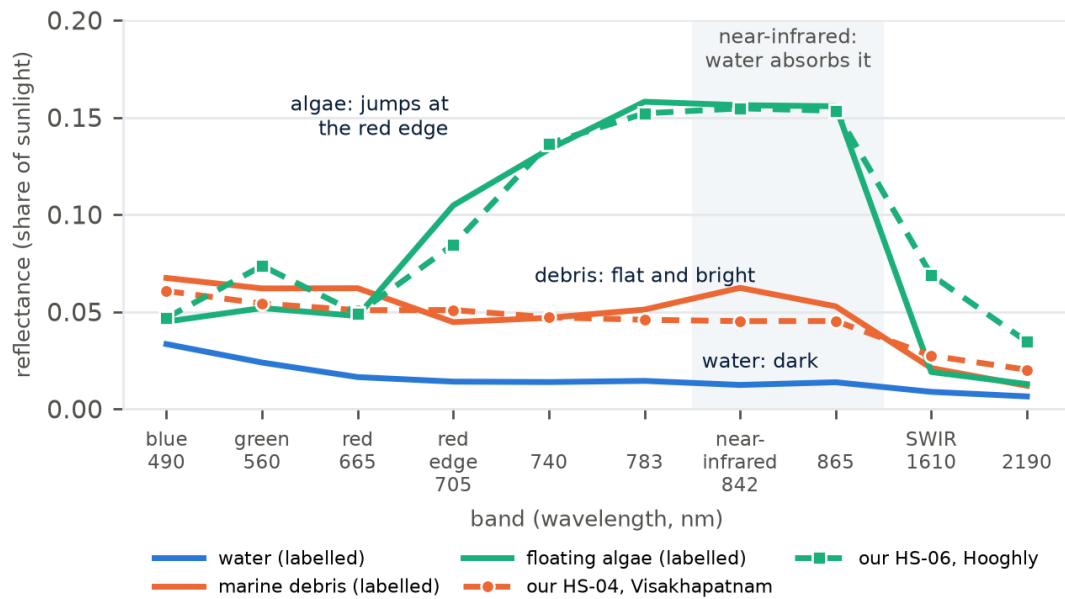
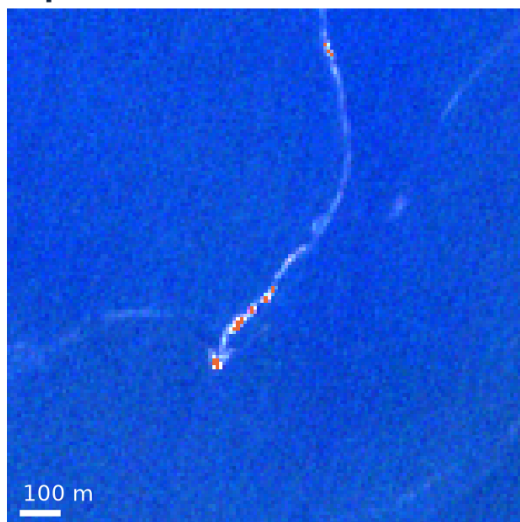


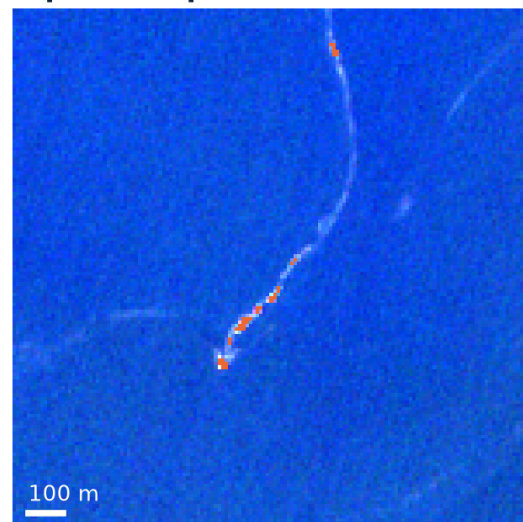
Figure 2c. Reflectance fingerprints. Solid: average of expert-labelled MARIDA pixels (water 86,877, debris 1,943, dense algae 870 pixels). Dashed: our Bay of Bengal hotspots.

Our Visakhapatnam hotspot follows the flat debris shape; the large Hooghly patch follows the vegetation shape (most likely floating water hyacinth), and the classifier labels them accordingly. This is why AquaTrace reports *potential* floating material with a type and a confidence: a spectrum can rule things in or out, but only a boat or drone can confirm plastic. Independent field experiments support the approach: plastic targets placed at sea were detected in Sentinel-2 imagery (Topouzelis et al., 2019), and FDI-based classification separated plastics from algae, driftwood and foam in several coastal waters (Biermann et al., 2020).

Experts' labels



AquaTrace prediction



orange = marine debris · 20 of 20 found · 0 of 69 water pixels flagged

Figure 2d. A MARIDA test scene from the Gulf of Honduras (15.8°N, 88.1°W), never used in training: the model found 20 of 20 expert-labelled debris pixels and flagged 0 of 69 labelled water pixels. Across the whole test split it finds 89% of the labelled debris.

Model selection, round 1 (validation split)

Features	Model	Labels	Accuracy	Macro F1	Debris F1	Floating F1	Train (s)
pixel	RF	equal	80.3%	0.575	0.755	0.395	12
pixel	RF	confidence	80.6%	0.579	0.794	0.405	12
pixel	ExtraTrees	equal	79.1%	0.570	0.788	0.406	4
pixel	ExtraTrees	confidence	78.6%	0.572	0.791	0.400	4
pixel	HistGB	equal	81.3%	0.558	0.654	0.375	151
pixel	HistGB	confidence	81.5%	0.564	0.672	0.370	118
pixel+texture	RF	equal	88.2%	0.701	0.822	0.841	17
pixel+texture	RF	confidence	88.3%	0.704	0.827	0.842	19
pixel+texture	ExtraTrees ✓ chosen	equal	88.1%	0.689	0.842	0.860	5
pixel+texture	ExtraTrees	confidence	88.1%	0.687	0.839	0.851	5
pixel+texture	HistGB	equal	89.8%	0.693	0.775	0.812	149
pixel+texture	HistGB	confidence	90.0%	0.700	0.768	0.808	173

Texture features gave the largest gain (about 8 points of accuracy; floating-material F1 from ≈ 0.40 to ≈ 0.85). Gradient boosting reached the highest validation accuracy (90.0%) but found less debris (F1 0.77 vs 0.84), so Extra Trees was chosen: the system optimises for the class that matters. Confidence weighting did not help Extra Trees.

Model selection, round 2 (validation split)

Candidate (validation split)	Accuracy	Macro F1	Debris F1	Floating F1
Extra Trees, features v1 (round-1 model)	88.3%	0.691	0.846	0.861
Extra Trees, features v2	89.6%	0.704	0.843	0.874
Extra Trees v2, 100,000 px per class	89.2%	0.698	0.845	0.879
Extra Trees v2, all pixels	89.0%	0.697	0.846	0.877
Gradient boosting (GB), features v2	93.9%	0.726	0.799	0.832
Soft vote 0.5 ET + 0.5 GB	94.0%	0.732	0.814	0.843
Soft vote 0.7 ET + 0.3 GB	93.9%	0.734	0.820	0.852
Extra Trees v2, debris if $P(\text{debris}) \geq 0.25$	89.6%	0.699	0.800	0.858
Extra Trees v2, debris if $P(\text{debris}) \geq 0.35$	89.6%	0.703	0.837	0.872
Two experts: ET decides floating, vote 0.0 ET + 1.0 GB	94.0%	0.740	0.843	0.874
Two experts: ET decides floating, vote 0.3 ET + 0.7 GB	94.1%	0.745	0.843	0.874
Two experts: ET decides floating, vote 0.5 ET + 0.5 GB ✓ chosen	94.1%	0.747	0.843	0.874
Two experts: ET decides floating, vote 0.7 ET + 0.3 GB	94.0%	0.745	0.843	0.874

The extended texture features (v2) improved accuracy and floating-material F1. More training pixels and a lower debris threshold did not help. Gradient boosting again had the best overall accuracy but weaker debris detection, so the two were combined as experts: Extra Trees decides floating material, the vote names everything else. That kept Extra Trees' debris and floating-material F1 exactly and raised validation accuracy from 89.6% to 94.1%. On the test split 12 of 15 classes improved, most for Wakes, Mixed Water and Waves.

Why background-relative features. A model trained on raw reflectance scored higher on MARIDA itself (debris F1 0.75 with pixel features) but labelled ordinary Bay of Bengal water as clouds and

wakes, because MARIDA patches and Sentinel-2 L2A scenes are processed differently. Subtracting the local water background fixed this: on the Sentinel-2 scenes, normal water is classed as marine, turbid or sediment-laden water, and only 0.00–0.13% of water pixels are flagged as floating material.

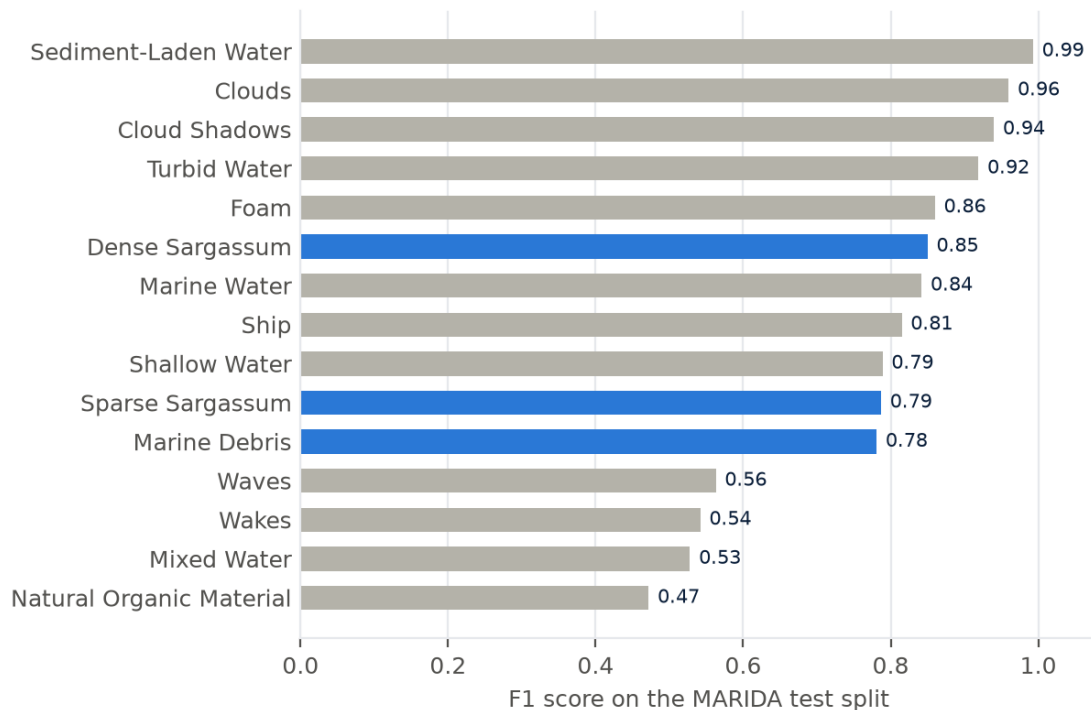


Figure 3. F1 score per class on the MARIDA test split. Blue: the floating-material classes used by AquaTrace.

Where the model goes wrong

True marine-debris pixels predicted as	Pixels	Pixels predicted as debris that were really	Pixels
Sparse Sargassum	16	Ship	49
Natural Organic Material	14	Clouds	28
Wakes	8	Natural Organic Material	21
Sediment-Laden Water	2	Wakes	20
		Waves	16

Correctly classified debris pixels: 338 of 381. The weakest classes (Foam, Natural Organic Material, Mixed Water, Wakes) have very few training pixels.

Most important features

Feature	Importance	Feature	Importance
dNDWI	0.049	dNDVI_m15	0.030
dNDVI_s15	0.036	dB2_s7	0.029
dNDVI_s7	0.035	dB2_m7	0.028
dNDVI_m7	0.032	dFDI_m15	0.027
dNDVI_m3	0.031	dFDI_m3	0.027

_m3/_m7/_m15 = mean in a $3 \times 3 / 7 \times 7 / 15 \times 15$ window, _s = standard deviation, _grad = edge strength (Extra Trees importances). Neighbourhood statistics of Δ NDVI, Δ FDI and the blue and near-infrared bands dominate, confirming that texture carries much of the signal.

6.2 Confidence calibration

A useful confidence must mean what it says: of all pixels given 80%, about 80% should be floating material. The reliability diagram compares stated confidence with observed frequency on the test split.

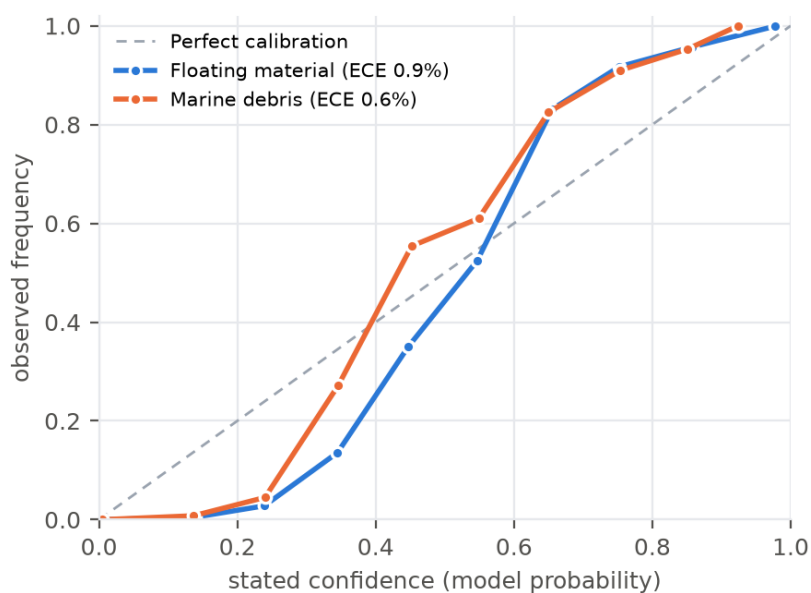


Figure 4. Reliability diagram (bins with at least 20 pixels).

Target	Expected calibration error (ECE)	Brier score	Positive pixels / total
Floating material	0.9%	0.0015	2,022 / 194,863
Marine debris	0.6%	0.0009	381 / 194,863

ECE is the average gap between stated confidence and observed frequency. The small values are helped by the many easy water pixels. The curve is S-shaped: pixels given 20–45% are floating material less often than stated (over-confident), while pixels given 60–90% are right more often than stated (under-confident: about 83% at a stated 66%). On Sentinel-2 AquaTrace only acts on pixels above 50%, so the part of the curve it uses errs on the safe side.

6.3 Drift forecast: validation against real buoys

Design. For every buoy and every day at 00:00 UTC with a full 48-hour track ahead, the model released particles at the buoy's true position and forecast its position at +24 h and +48 h. Drift settings were tuned on January 2025 (148 cases) and are scored here, once, on February 2025 (87 cases, 5 drifters, 2025-02-01 to 2025-02-24), which was not used for tuning. The error is the great-circle distance between the forecast ensemble mean and where the buoy really went. Two references: **persistence** (the buoy stays where it was seen, which is what a single satellite snapshot implies) and **currents only** (the same model without wind).

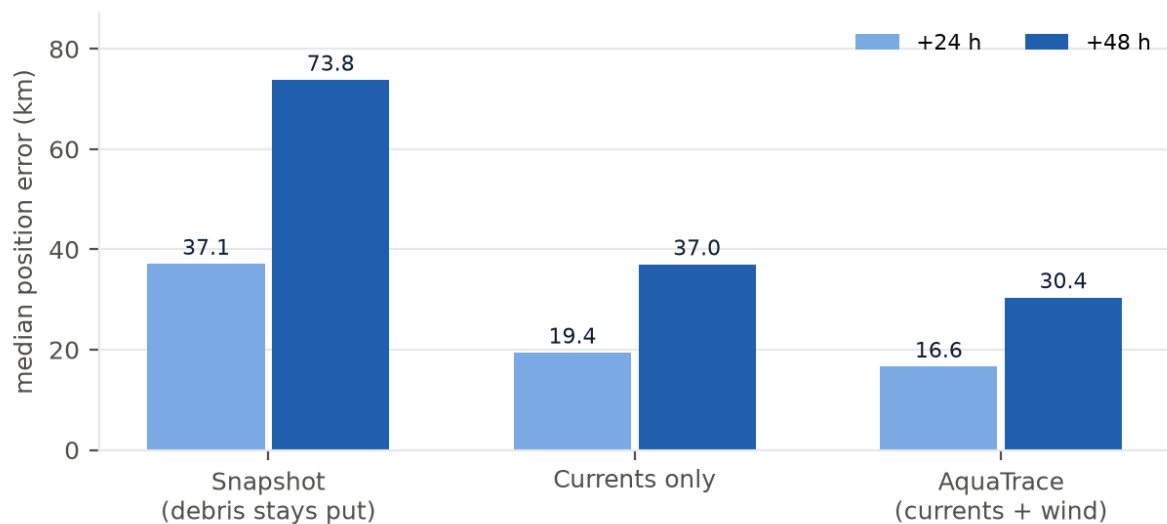


Figure 5. Median forecast error with NASA OSCAR currents and MERRA-2 wind (87 forecasts).

Method (NASA OSCAR + MERRA-2)	+24 h median	+24 h mean	+24 h 90th pct	+48 h median	+48 h mean	+48 h 90th pct
Snapshot (debris stays put)	37.1 km	41.2 km	69.7 km	73.8 km	80.2 km	122.6 km
Currents only	19.4 km	22.0 km	36.9 km	37.0 km	42.1 km	76.4 km
AquaTrace: currents + wind	16.6 km	18.6 km	30.6 km	30.4 km	33.8 km	57.2 km

Skill over a snapshot ($1 - \text{model error} / \text{persistence error}$): 55% at +24 h and 59% at +48 h. Adding wind to currents cut the 48-hour median error from 37.0 km to 30.4 km, direct evidence for the wind + current fusion in the project's design. For a cleanup boat, a 30 km search radius instead of 74 km is about 6 times less sea to search.

Is the uncertainty cone honest?

The map draws a cone that should contain 90% of outcomes. We checked whether the real buoy ended up inside it. With diffusion alone the cone was far too narrow: it held the buoy only 6% of the time at +24 h and 5% at +48 h. Adding a per-particle current error ($\sigma = 0.16$ m/s, chosen on January) raised this to 79% and 86% on the held-out month, close to the 90% target. Uncertainty in the ensemble is now a measured quantity, not a decoration.

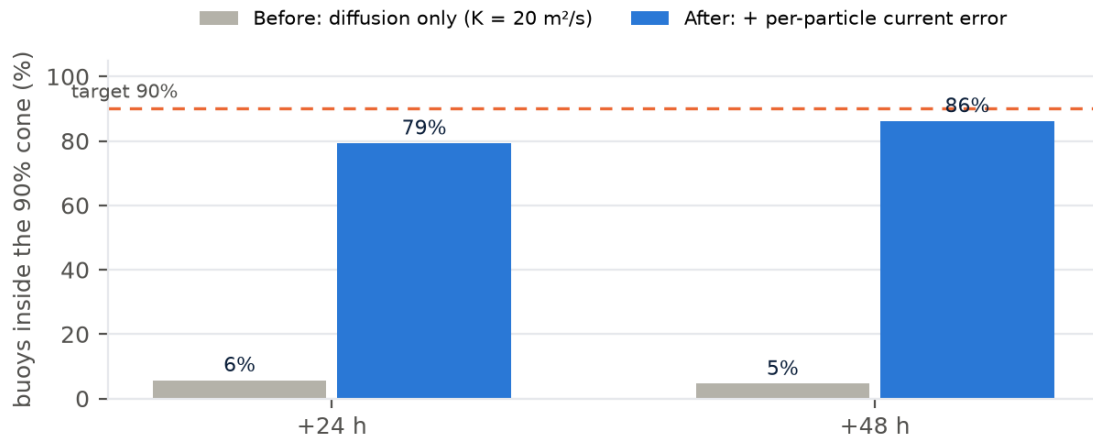


Figure 6. Share of real buoys inside the 90% cone on the held-out month (February 2025).

Setting (held-out month)	+24 h median	+48 h median	Inside cone +24 h	Inside cone +48 h	Cone area +48 h
Before: 1–3% wind, diffusion only	16.3 km	29.3 km	5.7%	4.6%	184 km ²
After: 2–3% wind + current error	16.6 km	30.4 km	79.3%	86.2%	9,713 km ²

The median position error barely changed (about 1 km, within the noise of five buoys): it is set by the daily, 25 km resolution of the current data. Scaling the currents up or down made it worse. What changed is that the cone now tells a cleanup crew honestly how wide to search.

Effect of the current and wind source

Forcing	Window	Cases	+24 h median	+48 h median	Skill +48 h
NASA OSCAR v2.0 + MERRA-2	2025-02-01 – 2025-02-24	87	16.6 km	30.4 km	59%
Open-Meteo (Mercator SMOC + ERA5)	2025-01-25 – 2025-02-08	65	24.3 km	43.0 km	45%

The open fallback, sampled on a coarser 0.5° grid, was less accurate. NASA forcing is preferred whenever it is available; live runs must use forecasts, because OSCAR and MERRA-2 arrive days to weeks late.

Windage calibration

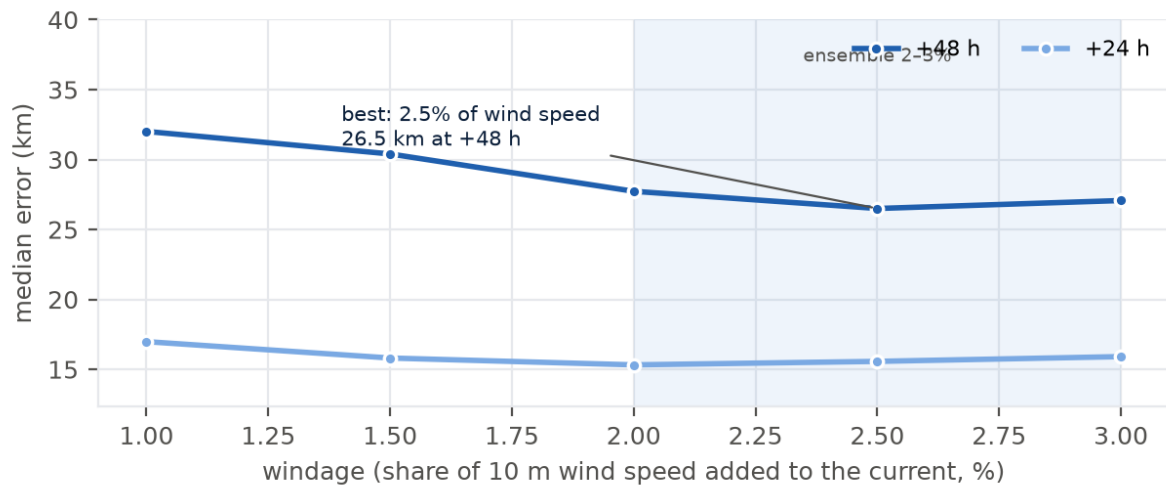


Figure 7. Median buoy error on the January tuning month as a function of windage, the share of 10 m wind speed added to the current.

On the January buoys the error was lowest at 2.5% of wind speed and rose for less or more wind. The operational ensemble samples 2–3% per particle around this optimum, so the spread also expresses uncertainty about debris type. The undrogued buoys float at the surface, the closest available proxy for debris, but their windage is not identical to that of plastic items.

7. Results from pipeline runs

Three runs are included with the project: a live run issued 30 September 2026 19:00 UTC and two replays (14 March 2026 with all NASA datasets, and 5 March 2026 with Sentinel-2 detection). Runtimes are with downloads already cached.

Run	Detection	Currents / wind	Scenes	Hotspots	High / Med / Low	Runtime
Live, 30 Sep 2026	NASA PACE OCI + ESA Sentinel-2	OSCAR + MERRA-2	6	25	0 / 1 / 24	66 s
Replay, 14 Mar 2026	NASA PACE OCI + ESA Sentinel-2	OSCAR + MERRA-2	5	17	5 / 2 / 10	47 s
Replay, 5 Mar 2026	ESA Sentinel-2	OSCAR + MERRA-2	5	13	3 / 9 / 1	42 s

Run	Sensor / site	Observed (UTC)	Usable water pixels	Candidate pixels	Hotspots	Cloud
Live, 30 Sep	PACE	2026-09-28 06:55	236,862	6,177	10	82.2%
Live, 30 Sep	PACE	2026-09-29 07:29	214,464	6,014	10	69.0%
Live, 30 Sep	PACE	2026-09-30 06:26	47,598	317	10	95.0%
Live, 30 Sep	S2 · Chennai coast	2026-09-26 05:15	1,569,990	1	0	–
Live, 30 Sep	S2 · Hooghly estuary / Sagar Island	2026-09-27 04:52	1,347,600	2,990	3	–

Run	Sensor / site	Observed (UTC)	Usable water pixels	Candidate pixels	Hotspots	Cloud
Live, 30 Sep	S2 · Visakhapatnam coast	2026-09-28 05:03	1,730,680	23	3	–
Replay, 14 Mar	PACE	2026-03-14 06:59	584,107	16,875	10	56.3%
Replay, 14 Mar	S2 · Chennai coast	2026-03-15 05:15	1,486,841	5	1	–
Replay, 14 Mar	S2 · Visakhapatnam coast	2026-03-12 05:03	1,735,612	31	3	–
Replay, 14 Mar	S2 · Cox's Bazar coast	2026-03-13 04:32	1,725,007	233	3	–
Replay, 14 Mar	S2 · Port Blair (Andaman)	2026-03-15 04:25	874,781	2	0	–

The live run reflects post-monsoon conditions: the 30 September pass was 95% cloud, so most candidates come from the 28–29 September passes and were nowcast 36–86 hours forward. Live tiers (High / Medium / Low): 0 / 1 / 24, an honest result for a cloudy period: old detections carry wide cones, which lowers their certainty.

7.1 Example: a hotspot off Chennai (replay, 14 March 2026)

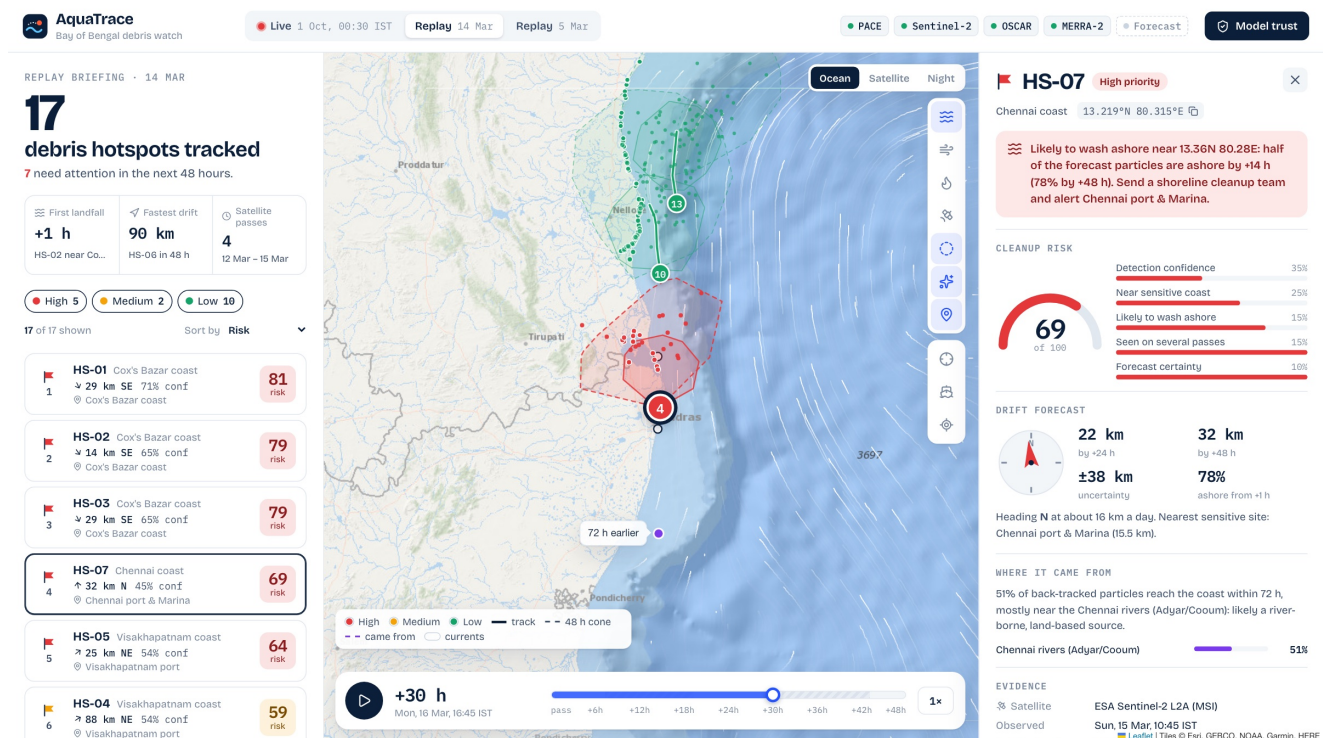


Figure 8. Hotspot HS-07 at +30 h: the forecast cone runs north along the coast, the purple back-track shows where it came from, and the drawer explains the recommendation.

HS-07 was detected by Sentinel-2 on 2026-03-15 with confidence 0.45. The forecast takes it 32 km north in 48 h, with 78% of particles reaching the shore (first after 1 h, half by +14 h). Risk 69 (High). Recommended action: “Likely to wash ashore near 13.36N 80.28E: half of the forecast particles are ashore by +14 h (78% by +48 h). Send a shoreline cleanup team and alert Chennai port & Marina.” Source tracking: “51% of back-tracked particles reach the coast within 72 h, mostly near the Chennai rivers (Adyar/Cooum): likely a river-borne, land-based source.”

7.2 Example: live map and cleanup route (30 September 2026)

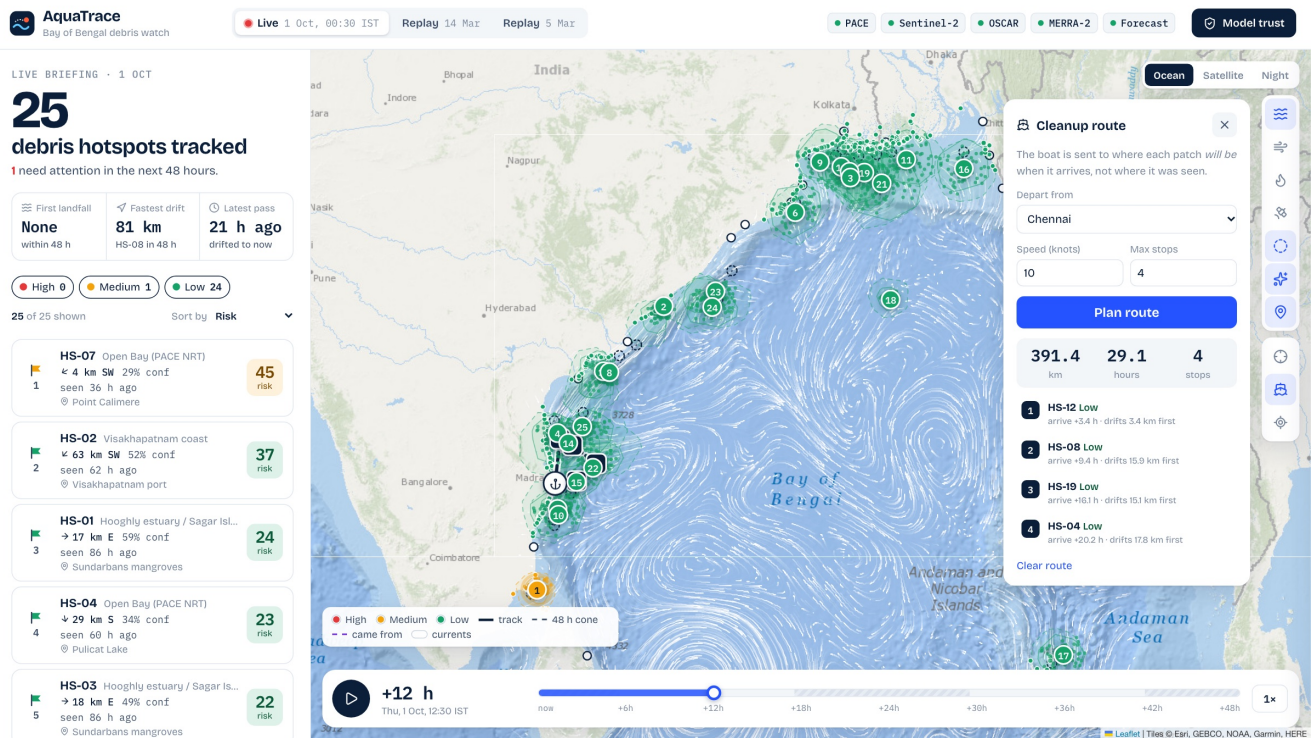


Figure 9. Live run with a cleanup route from Chennai: 4 stops, 391.4 km, 29.1 h at 10 knots.

Stop	Hotspot	Tier	Arrive	Leg	Drift before the boat arrives
1	HS-12	Low	+3.4 h	62.5 km	3.4 km
2	HS-08	Low	+9.4 h	74.9 km	15.9 km
3	HS-19	Low	+16.1 h	85.8 km	15.1 km
4	HS-04	Low	+20.2 h	40.2 km	17.8 km

A second example from the replay: departing Port Blair, 2 stops, 407.0 km, 26.0 h. In both cases patches move up to 21 km before the boat arrives, which is why the planner targets the forecast position, not the last sighting. High-risk patches that are forecast to wash ashore go to shoreline teams instead.

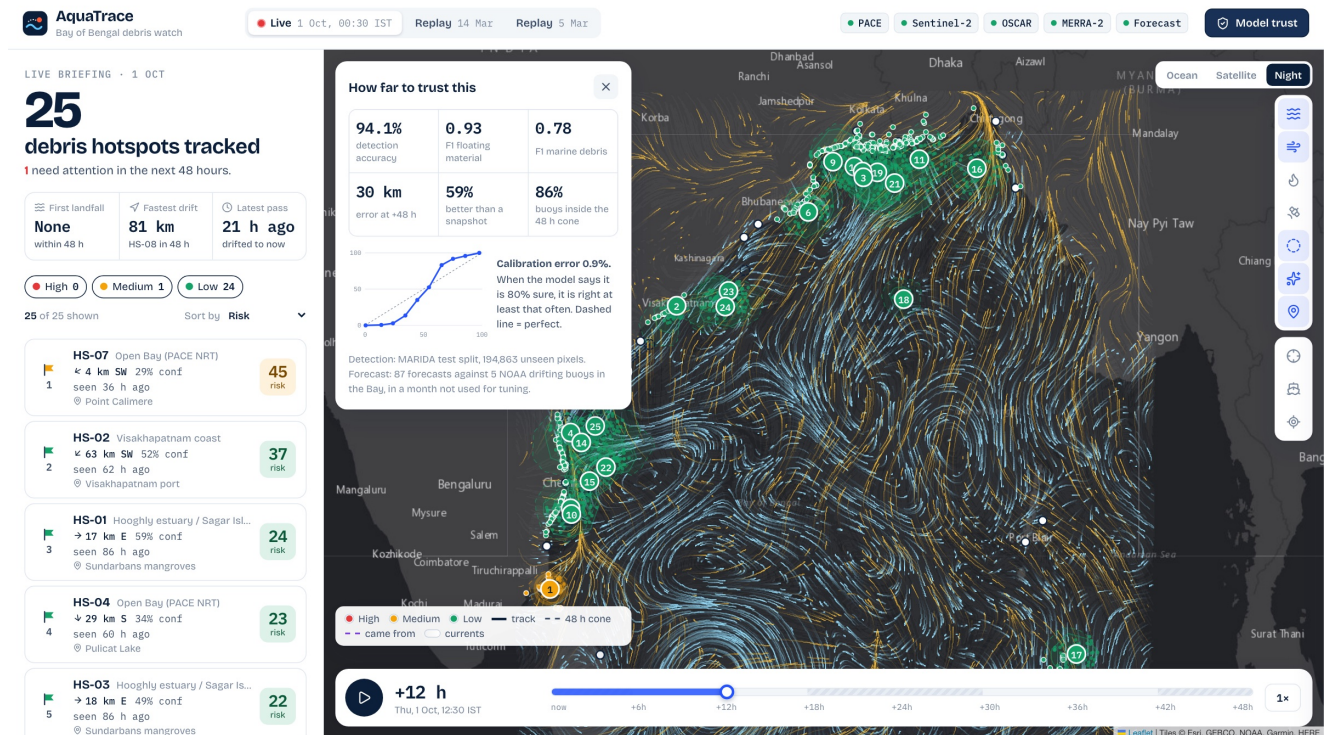


Figure 10. Night basemap with animated currents (cyan) and wind (amber), the debris-signal layer, and the Model trust panel.

8. Software and architecture

8.1 Technology stack

Layer	Technology
Language / runtime	Python 3.14 (backend, pipeline), JavaScript (React) for the dashboard
Scientific	NumPy, SciPy, pandas, xarray, netCDF4 / h5netcdf, rasterio, pyproj, GeoPandas, Shapely
Machine learning	scikit-learn 1.9 (Extra Trees, histogram gradient boosting), joblib
Data access	earthaccess (NASA Earthdata), pystac-client (Earth Search STAC), requests (ERDDAP, Open-Meteo, OPeNDAP)
API	FastAPI + uvicorn
Dashboard	React 19, Vite 8, Leaflet 1.9 / react-leaflet 5, lucide-react icons, custom canvas layers for animated flow and particles
Reporting	matplotlib, ReportLab (this report), python-pptx (pitch deck)

8.2 Backend API

There is no database. The pipeline writes files; the API reads and serves them, and computes what-if forecasts and routes on demand.

Endpoint	Returns
GET /api/runs	list of runs (from outputs/runs/index.json)
GET /api/runs/{id}	one run: hotspots, forecasts, overlays, sources, current/wind samples
GET /api/runs/{id}/files/{path}	satellite true-colour and debris-anomaly PNG overlays
GET /api/metrics	classifier metrics, calibration, buoy validation
POST /api/forecast	what-if: 48 h ensemble forecast and risk score for any sea point (not stored)
GET /api/runs/{id}/route	moving-target cleanup route from a chosen port (not stored)
GET /api/ports	port anchorages for the route planner
GET /	the dashboard (frontend/dist)

8.3 Where data is stored

Folder	Contents	Size
data/raw/	downloaded inputs: PACE granules, OSCAR, MERRA-2 subsets, MARIDA, NOAA drifters, Natural Earth	≈10 GB
data/processed/	caches: current/wind grids (.nc), Sentinel-2 scenes (.npz)	≈0.5 GB
models/	trained classifier (marida_rf.joblib, ~630 MB), metrics, calibration, experiments (.json)	≈0.65 GB
outputs/runs/<id>/	run.json (all results of one run) + overlay PNGs; index.json lists runs	≈14 MB
outputs/	drift_validation.json (held-out month), drift_tuning.json	small
~/.netrc (home folder)	the user's Earthdata login, deliberately outside the project	–

8.4 Dashboard

The dashboard is designed as a maritime operations chart. A bathymetric ocean basemap carries animated ocean currents, and risk tiers use the international signal-flag colours (red, yellow, green). Main elements:

- **Top bar:** switch between the live run and replays, see which datasets the run used, open Model trust.
- **Briefing:** headline count, first landfall, fastest drift, latest pass; tier filters, sorting, and a ranked list linked to the map by hover.
- **Map:** pulsing priority pins, 48 h tracks and uncertainty cones, drifting particles, nowcast trails, back-tracks, routes.
- **Detail drawer:** recommended action, risk gauge and breakdown, drift compass, source, evidence, copyable coordinates.
- **Time bar:** smooth 48 h playback (Space), arrow-key scrubbing, 1×/2×/4× speed, night bands (IST).
- **Toolbar:** Ocean / Satellite / Night basemaps, seven layers, what-if simulator, route planner. Views are shareable as links.

8.5 Performance

Task	Time
Detect one PACE granule (1.8 million pixels in the Bay window)	≈13 s
Replay run with cached data (17 hotspots, forecasts, back-tracks)	≈47 s
Live run (3 PACE + 3 Sentinel-2 scenes, forecast forcing; PACE already downloaded)	≈1 min
Train the final classifier (500 Extra Trees + 400 boosting rounds)	≈134 s
What-if forecast (300 particles, 48 h)	≈1 s

9. How to run it

```

./setup.sh          # one-time: Python environment (+ dashboard build if npm exists)
./run_demo.sh       # API + dashboard on http://localhost:8000

cd backend
./venv/bin/python -m aquatrace.pipeline --live          # near-real-time run
./venv/bin/python -m aquatrace.pipeline --date 2026-03-14 # replay with NASA data
./venv/bin/python -m aquatrace.marida                  # retrain the classifier
./venv/bin/python -m aquatrace.experiments              # model selection (validation)
./venv/bin/python -m aquatrace.calibration              # reliability / ECE
./venv/bin/python -m aquatrace.experiments_v2           # round-2 selection
./venv/bin/python -m aquatrace.tune_drift               # tune drift (Jan buoys)
./venv/bin/python -m aquatrace.validate --holdout       # score on Feb buoys

```

- **Earthdata login** (free) is needed for PACE, OSCAR and MERRA-2. Approve *NASA GESDISC DATA ARCHIVE* in the Earthdata profile for MERRA-2, then save the login once with earthaccess (stored in ~/.netrc).
- **Without a login**, runs fall back to Sentinel-2 and Open-Meteo automatically, and the dashboard reports which sources were used.
- **Sharing**: the full project zip (≈475 MB) contains code, saved runs, cached forcing and the trained model. The dashboard can be updated on its own with a ~200 KB frontend package.
- **Shareable views**: e.g. ?run=20260314_pace&select=HS-07&hour=24 or ?basemap=dark&show=wind&trust=1.

10. Limitations and responsible use

- **Not a live video of debris**. PACE passes once a day at about 1.2 km, Sentinel-2 every few days near the coast, and clouds block both. AquaTrace shows *potential* floating-material hotspots with a confidence, carried forward by a validated drift model.
- **Sub-pixel detection on PACE**. Individual items are far smaller than a 1.2 km pixel; PACE flags regions where floating material (debris, algae, foam) changes the spectrum. Sentinel-2 is the higher-resolution check.
- **Training data are not from India**. MARIDA scenes come from other seas; detection scores are on MARIDA's test split. Local ground truth (drones, beach surveys, fishers' reports) is needed to measure accuracy in the Bay.
- **Reanalysis latency**. OSCAR and MERRA-2 arrive days to weeks late, so NASA runs are replays; live runs use forecast currents and wind, which were less accurate in our comparison.
- **Validation proxy**. Buoys test the current and wind field and windage of surface drifters, not of plastic items; five drifters over eight weeks is a modest sample (87 held-out forecasts). The current error is isotropic, so for patches within a few kilometres of the shore it may overstate how many particles land.

- **Approximate reference data.** Sensitive sites, river mouths and port anchorages are hand-entered with approximate coordinates; routes are straight legs with simple detours, not navigational charts.
- **Decision support, not automation.** Recommendations should be confirmed by people before deploying vessels.

11. Roadmap

Phase	Next steps
1 · Pilot (done)	Bay of Bengal prototype: detection, 48 h forecast, risk, routes, live mode, validation
2 · Validate on the ground	Fishers' and citizens' geotagged reports and drone surveys as local labels; retrain and re-score; add Sentinel-1 radar for cloudy days
3 · Expand across India	Arabian Sea and Lakshadweep; integrate INCOIS ocean forecasts; automated daily runs with alerts to coastal authorities
4 · Scale beyond borders	Full hyperspectral PACE unmixing; cyclone scenario mode; spatial database (PostGIS) and cloud storage for operations

12. References

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Appendix A. Key parameters

Parameter	Value	Where
Pilot region	4°–23°N, 79°–96°E	config.py
Particles per hotspot / time step / horizon	300 / 1 h (RK4) / 48 h	config.py, drift.py
Windage / diffusivity / current error	2–3% (uniform per particle) / 20 m ² /s / $\sigma = 0.16$ m/s ($\pm$ pairs)	config.py
Anomaly threshold	$z \geq 3.5$ (robust, MAD-based)	detect.py
Background block	41 px (PACE) / 101 px (Sentinel-2 at 20 m)	detect.py
Shoreline strip excluded	2.5 km (PACE) / 0.4 km (Sentinel-2)	detect.py
Cloud buffer	3 px (PACE) / 5 px (Sentinel-2)	detect.py, sentinel2.py
Sentinel-2 window / resolution / search	30 × 30 km / 20 m / ± 6 days, < 20% cloud	sentinel2.py
Classifier	Extra Trees (500 trees) decides floating; 50/50 vote with gradient boosting (400 rounds) names the rest; $\leq 40,000$ px per class	marida.py
Risk weights	confidence 35, proximity 25, beaching 15, recurrence 15, certainty 10	risk.py
Tiers	High ≥ 60 , Medium ≥ 40	risk.py
Back-track / recurrence radius	72 h / 15 km	source.py, pipeline.py
Route defaults	10 kn, 2 h per stop, 450 km range, detour offsets 10–110 km	route.py
Live look-back	3 days, best 3 PACE NRT passes	pipeline.py

Appendix B. Project files

Path	Purpose
backend/aquatrace/sources/	nasa.py (PACE, OSCAR, MERRA-2), sentinel2.py, openmeteo.py, coast.py
backend/aquatrace/forcing.py	common velocity field and interpolation
backend/aquatrace/detect.py	band synthesis, masks, indices, texture features, anomaly, hotspots, overlays
backend/aquatrace/marida.py, experiments.py, experiments_v2.py, calibration.py	classifier training, model selection (two rounds), calibration
backend/aquatrace/drift.py, source.py	Lagrangian forecast (forward and backward), source attribution
backend/aquatrace/risk.py, route.py	risk score and actions; moving-target route planner
backend/aquatrace/validate.py, tune_drift.py	buoy validation; drift tuning on one month, scoring on the next
backend/aquatrace/pipeline.py, api.py	replay and live runs; FastAPI service
frontend/src/	dashboard: App, components (briefing, drawer, timeline, toolbar), map (flow and particle layers)
README.md, CLAUDE.md, DEMO.md	setup and results; developer guide; demo script and judge Q&A

Appendix C. Glossary

Term	Meaning
Hotspot	a group of pixels with a strong floating-material signal; a <i>potential</i> debris patch
FDI	Floating Debris Index: near-infrared reflectance above a red-edge/SWIR baseline
Lagrangian model	tracks individual particles as they are carried by currents and wind
Ensemble	many particles with slightly different properties; their spread is the uncertainty
Windage	the share of wind speed that pushes a floating object in addition to the current
Nowcast	estimate of where a past detection is now
Replay (hindcast)	a run on a past date that can be checked against what happened
Precision / recall / F1	share of flagged pixels that are right / share of true pixels found / their harmonic mean
ECE	expected calibration error: gap between stated confidence and observed frequency
Persistence	the baseline forecast that nothing moves (what a single snapshot implies)
Cone coverage	share of real outcomes that fall inside the forecast's 90% uncertainty cone (ideal: 90%)
RK4	fourth-order Runge-Kutta, an accurate way to step a particle through a velocity field